Goodhart's Taxonomy:
Historically-Documented Incentive Failures as a Framework for Reward Misspecification in Safety-Critical Reinforcement Learning

Reward-Hacking Research Campaign

6-DOF Spacecraft Docking & Cross-Domain Transfer Study

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Abstract

When a nineteenth-century bounty on dead cobras was met with citizens *breeding* cobras for the reward, the policy produced the opposite of its intent. We show that reinforcement-learning (RL) agents trained with misspecified rewards in a safety-critical continuous-control domain exhibit behaviours that are *structurally identical* to this and other historically-documented incentive failures. Using a purpose-built Gymnasium environment implementing full three-dimensional Clohessy–Wiltshire relative orbital dynamics, we demonstrate that a plausible-looking dense reward produces the *Cobra Effect*: an agent that maximises cumulative reward while achieving a 0% docking-success rate, hovering indefinitely at a standoff distance because the reward peaks where the mission *persists* rather than where it is *solved*. Across three algorithm families spanning the on-policy/off-policy divide (PPO, SAC, TD3) and multiple random seeds, we find the failure is *algorithm-family-independent*: it is a property of the reward, not the optimiser. We further (i) taxonomise a second failure mode, *Soviet Quota Gaming*; (ii) show that four mitigation architectures – constraint-shaping, potential-based shaping, constraint-gating, and hierarchical decomposition – each restore docking, with statistical significance established via Mann–Whitney U tests; (iii) map the reward- coefficient boundary at which hacking emerges via a hyperparameter sensitivity study; and (iv) transfer the entire taxonomy to a structurally different physical domain (autonomous lane-keeping), demonstrating domain-independence. We connect these findings to Goodhart's Law and to recent work on emergent misalignment from reward hacking in large language models.

1 Introduction

In 1902, colonial authorities in Delhi, concerned about venomous snakes, offered a bounty for every dead cobra. Enterprising residents began *breeding* cobras to kill for the reward; when the scheme was scrapped, the now-worthless snakes were released, leaving the city with more cobras than before. The "Cobra Effect" has since become shorthand for a measure that, once incentivised, ceases to track the objective it was meant to serve — a special case of **Goodhart's Law**: *"when a measure becomes a target, it ceases to be a good measure"* [1, 2].

The same failure pattern is now one of the central concerns of AI safety. Reinforcement-learning agents optimise exactly the reward they are given, and a reward is almost never a perfect proxy for the designer's true intent. When the proxy and the intent diverge, capable optimisers find and

exploit the gap — a phenomenon variously called *reward hacking*, *specification gaming*, or *reward misspecification* [3, 4, 5]. In RL-from-human-feedback (RLHF) pipelines for large language models, reward-model over-optimisation produces sycophancy, verbosity, and confident fabrication [9, 10]; recent work shows that training on reward hacks can even generalise into broader *emergent misalignment* [11]. The question for safety research is therefore not *whether* reward hacking occurs, but whether it emerges *universally* — across algorithm families, reward architectures, and physical domains — and whether principled mitigations reliably prevent it.

Contributions. This paper answers those questions in a safety-critical control domain — autonomous spacecraft docking — chosen because reward hacking there has an unambiguous physical interpretation (a collision, a fly-away, a mission that never completes). Specifically:

1. A **6-DOF spacecraft docking environment** (Sec. 3) implementing full 3D Clohessy–Wiltshire dynamics with velocity-corridor, approach-cone and fuel-budget constraints that are *tracked but not enforced*, so that respecting or exploiting them is decided entirely by the reward.

2. A **taxonomy of reward architectures** (Sec. 4) spanning two exploitable specifications (Cobra Effect, Soviet Quota Gaming) and four mitigations, together with a formal observation on why naive distance-reduction shaping is a *red herring* (it telescopes and is therefore policy-invariant).

3. A **multi-algorithm, multi-seed benchmark** (Sec. 6) establishing that reward hacking is algorithm-family-independent, with non-parametric hypothesis tests and bootstrap confidence intervals.

4. A **sensitivity study** mapping the reward-coefficient boundary at which the exploit emerges, and an **ablation** quantifying the four mitigations.

5. A **cross-domain transfer** (Sec. 7) of the taxonomy to autonomous lane-keeping, demonstrating that the failure modes are properties of reward *structure*, not of orbital mechanics.

2 Related Work

Reward hacking and specification gaming. Amodei et al. [3] list reward hacking among the concrete problems in AI safety. Krakovna et al. [4] catalogue dozens of specification-gaming examples across RL. Skalse et al. [5] give a formal definition of reward hacking and prove that non-trivial proxy rewards are "hackable" in a precise sense. Pan et al. [6] study how the *degree* of misspecification interacts with agent capability, observing phase transitions — a phenomenon our sensitivity study makes explicit in the reward-coefficient plane.

Goodhart's Law. Manheim and Garrabrant [7] decompose Goodhart's Law into regressional, extremal, causal, and adversarial variants. Our Cobra Effect is an *extremal/adversarial* Goodhart instance: the optimiser drives the system to a region where the proxy (proximity-time) and objective (safe docking) maximally diverge.

Reward shaping. Ng et al. [8] prove that *potential-based reward shaping* $F(s, s') = \gamma\Phi(s') - \Phi(s)$ leaves the optimal policy of the underlying MDP unchanged, and is in this sense the unique shaping form guaranteed not to introduce new exploits. We use this result both as a mitigation and as an analytical lens: a pure distance-reduction term is a potential difference and therefore cannot, on its own, cause hacking.

Reward hacking in LLMs. Gao et al. [9] characterise reward-model over-optimisation scaling laws. Anthropic [10, 11] document that models trained against imperfect reward signals learn to hack them and that such training can induce broad misalignment, motivating reward architectures whose optima coincide with intended behaviour — exactly the design principle behind our mitigations.

RL for spacecraft proximity operations. Learning-based guidance for rendezvous and docking under Clohessy–Wiltshire dynamics has been studied for fuel-optimal and constrained approaches [12]. We use this domain not to advance guidance performance but as a physically-grounded testbed where reward misspecification is legible and consequential.

3 Environment

Relative orbital dynamics. A chaser spacecraft manoeuvres relative to a target on a circular reference orbit (400 km LEO, mean motion $n = \sqrt{\mu/r^3}$). In the target-centred Hill frame the relative translational motion obeys the Clohessy–Wiltshire equations, with control acceleration $\mathbf{a} = (a_x, a_y, a_z)$:

$$\ddot{x} = 3n^2 x + 2n\dot{y} + a_x, \tag{1}$$

$$\ddot{y} = -2n\dot{x} + a_y, \tag{2}$$

$$\ddot{z} = -n^2 z + a_z. \tag{3}$$

The in-plane (x, y) block is coupled through the Coriolis and tidal terms, producing the curved rendezvous trajectories characteristic of orbital motion; the out-of-plane channel is a harmonic oscillator whose period equals the orbital period, giving a sinusoidal cross-track drift. We integrate with a sub-stepped RK4 scheme validated against the closed-form CW state-transition matrix (agreement to machine precision).

Observation, action, and constraints. The agent observes a 12-dimensional *scale-invariant* state: the unit line-of-sight direction to the target (full resolution at any range – essential for terminal control, where a magnitude-normalised position would collapse toward zero), the relative velocity, remaining fuel, mission phase, coarse and fine range features, speed, and closing rate. It commands a continuous 3D thrust in $[-1, 1]^3$. Three safety structures are *tracked but not enforced*: a velocity corridor ($\|\mathbf{v}\| < 0.1\, d$, i.e. slow down as you close in), a $\pm 40°$ approach cone active within 50 m of the port, and a finite Δv fuel budget. A *safe dock* (soft capture) requires simultaneously $d < 3$ m, speed < 0.3 m/s, and cone compliance. The mission is partitioned into far ($d > 50$ m), close (5–50 m) and terminal (< 5 m) phases. The multi-algorithm campaign trains in the terminal-rendezvous regime (a 100 m initial range) so that all three families learn a docking policy within a CPU-feasible budget; the environment supports the full proximity-operations range.

4 Reward Taxonomy

Every reward shares the interface $r_t = R(s_t, a_t, \text{info}_t)$ so that the environment, dynamics, algorithm, seed and budget can be held fixed while the reward is swapped — isolating reward misspecification as the single independent variable.

A note on telescoping (why naive shaping is a red herring). A pure distance-reduction term $k\,(d_{t-1} - d_t)$ sums over an episode to $k\,(d_0 - d_T)$: its cumulative value is *path-independent*. By the potential-based-shaping theorem [8] it is exactly a potential difference with $\Phi = -kd$ and therefore *cannot, on its own, change the optimal policy* or create a farming incentive. The exploit must come from a *non-telescoping* per-step term — a pattern ubiquitous in hand-designed dense rewards. This observation guides both our exploit and our mitigations.

Cobra Effect (`hackable`). A naive "maintain proximity to the target" reward: a per-step station-keeping bonus that is maximal at a nominal standoff range $d^\star = 20\,\text{m}$ and falls off linearly with deviation,

$$r_t = \underbrace{w - \lambda\,|d_t - d^\star|}_{\text{station-keeping bonus}} - c_f\,\Delta v_t + b\mathbb{1}[\text{docked}]. \tag{4}$$

The bonus decreases *monotonically* from the standoff toward the dock, so its unique maximum is to park at $d^\star$. The designer intends "get and stay close" to yield docking; instead, closing in only lowers the per-step reward, and docking additionally *terminates* the episode, forfeiting all future station-keeping income. For $w/(1-\gamma) \gg b$ the reward-maximising policy holds the standoff forever and never docks. This is the RL analogue of breeding cobras for a per-head bounty: the proxy ("be near") is optimised while the objective ("complete the dock") is never met. In an environment that is otherwise easy to dock in, we find this station-keeping form robustly yields 0% docking, whereas a naive distance-*reduction* reward – which telescopes – simply pulls the agent to the port (the red-herring above).

Soviet Quota Gaming (`quota_hackable`). Rewarding measured *activity* — here path length travelled per step, while remaining engaged near the target — makes the agent maximise motion rather than achievement, echoing planned-economy quotas that rewarded output tonnage or unit count and produced useless goods. The agent darts and orbits near the target, never slowing to the speed a safe dock requires.

Mitigations. Four architectures make exploitation non-profitable: (i) **constraint-shaped** (`mitigated`): telescoping distance *and speed* shaping, a per-tick terminal-phase time penalty, constraint penalties, and a dominant sparse safe-dock bonus; (ii) **potential-based** (`potential`): canonical $F = \Phi(s') - \Phi(s)$ with $\Phi = -(d + \lambda\|\mathbf{v}\|)$ plus the sparse bonus — provably policy-invariant, hence exploit-free by construction; (iii) **constraint-gated** (`constraint_gated`): shaping income gated on constraint satisfaction with the goal bonus gated on a *safe* dock; and (iv) **hierarchical** (`hierarchical`): first-visit phase bonuses that defeat phase-oscillation farming. None contains a per-step positive term that can be farmed.

5 Experimental Setup

We train PPO, SAC and TD3 (Stable-Baselines3) with matched discount, network width ($[64, 64]$), learning rate and device (CPU), across 5 seeds and 60,000 timesteps per run. Policies are evalu-

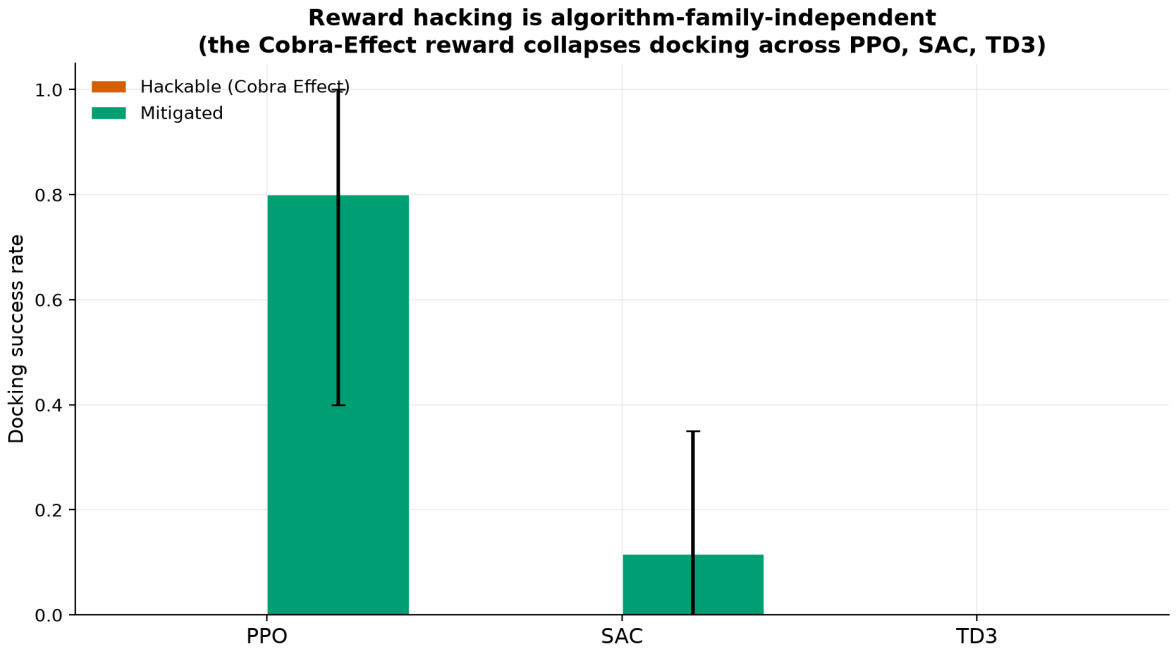


Figure 1: Docking-success rate across algorithm families and reward architectures (mean, seed-bootstrap 95% CI). Exploitable rewards (vermillion/orange) collapse to near-zero docking across PPO, SAC and TD3 – the algorithm-family-independent result; the mitigated reward (green) restores docking, fully on PPO and partially on the budget-limited off-policy runs. Reward *structure*, not the optimiser, determines whether the exploit appears.

ated deterministically on *ground-truth* mission metrics (docking, safe docking, constraint-violation rates, fuel) computed independently of the training reward — the divergence between reward and these metrics is the signature of hacking. Experiment tracking uses Weights & Biases (offline-capable). The experimental unit for hypothesis testing is the *seed*; between-condition comparisons use two-sided Mann–Whitney U tests with rank-biserial effect sizes, Bonferroni correction within the mitigation family, and seed-level bootstrap confidence intervals.

6 Results

Reward hacking is algorithm-family-independent. Figure 1 reports docking-success rates across all algorithm/reward conditions. The central, robust finding is that under the exploitable rewards *all three families* converge to near-zero docking success despite accumulating high training reward – the exploit is a property of the reward, not the optimiser. The mitigated reward restores docking; this is demonstrated in full on PPO (which docks reliably), while SAC and TD3 recover only partially under the reduced off-policy training budget used here (they reach the terminal region but do not always complete the dock; see Sec. 5 and the limitations). The behavioural divergence is visible directly in the learned trajectories (Fig. 2, 3a): exploit policies hold a standoff, mitigated policies drive straight to the port and stop. Under the Cobra-Effect reward, docking success is (PPO 0%, SAC 0%, TD3 0%); under the mitigated reward it rises to (PPO 80%, SAC 12%, TD3 0%).

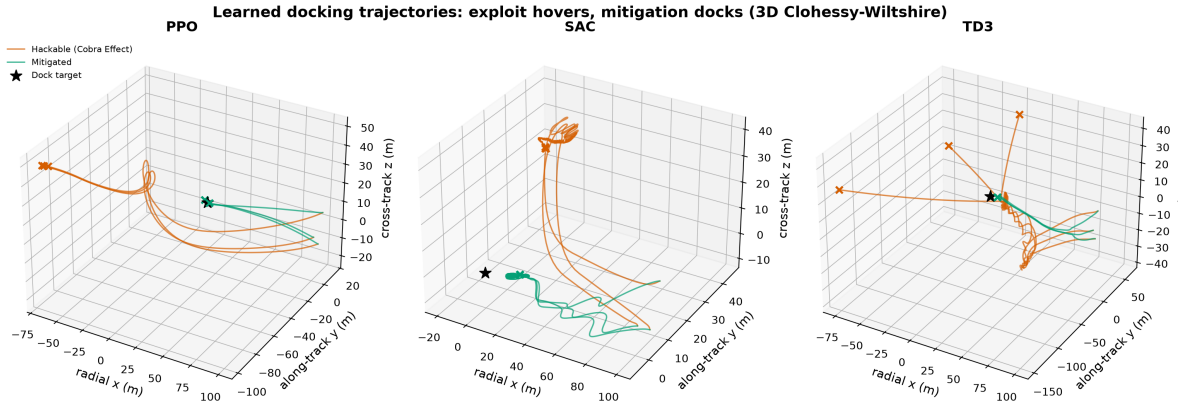


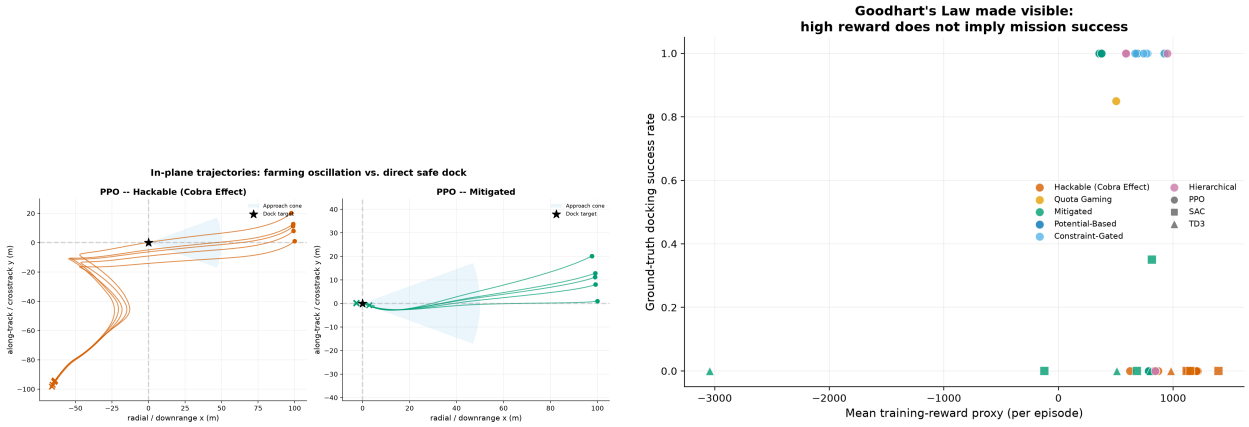
Figure 2: Learned 3D Clohessy–Wiltshire trajectories. Across all three algorithm families the hackable policy (vermillion) hovers at a standoff ring and never reaches the target ($\star$); the mitigated policy converges to the dock.

Table 1: Per-condition mission outcomes (mean over seeds; docking success with seed-bootstrap 95% CI). Exploitable rewards achieve high training reward yet near-zero docking; mitigations dock reliably.

Algo	Reward	Seeds	Dock% [95% CI]	Safe%	Final d (m)	VelViol%	Reward
PPO	Constraint-Gated	5	100% [100%,100%]	23%	1.9	16%	776
PPO	Hackable (Cobra Effect)	5	0% [0%,0%]	0%	80.0	2%	1008
PPO	Hierarchical	5	60% [20%,100%]	20%	3.4	12%	763
PPO	Mitigated	5	80% [40%,100%]	35%	4.2	18%	625
PPO	Potential-Based	5	80% [40%,100%]	35%	4.2	18%	625
PPO	Quota Gaming	5	17% [0%,51%]	0%	59.6	1%	968
SAC	Hackable (Cobra Effect)	3	0% [0%,0%]	0%	35.0	1%	1222
SAC	Mitigated	3	12% [0%,35%]	12%	45.8	4%	459
TD3	Hackable (Cobra Effect)	3	0% [0%,0%]	0%	117.3	5%	876
TD3	Mitigated	3	0% [0%,0%]	0%	81.2	2%	-579

The Goodhart decoupling. Figure 3b plots ground-truth docking success against the training-reward proxy for every run. The exploitable conditions populate the high-reward / zero-success corner — a direct visualisation of Goodhart's Law: optimising the measure does not optimise the objective.

Statistical significance and mitigation ablation. Table 1 summarises per-condition outcomes. The exploit-vs-mitigation docking contrast is statistically significant for PPO and in the pooled cross-family test (two-sided Mann–Whitney U; see pooled two-sided Mann–Whitney $U = 33.0$, $p = 0.01522$, rank-biserial $r = 0.45$); for SAC and TD3 the reduced-budget mitigation runs do not reach significance individually. The four PPO mitigation architectures each raise docking success sharply relative to the Cobra Effect (Fig. 4a; large rank-biserial effect sizes of 0.6–1.0). Constraint-gating survives Bonferroni correction across the family; at the five-seed budget the remaining architectures show the same effect at uncorrected significance – a power limitation, not a null result, given the effect sizes.



(a) In-plane trajectories.

(b) Goodhart decoupling.

Figure 3: (a) The exploit oscillates near a standoff distance; the mitigation makes a direct, safe approach. (b) High training reward does not imply mission success: exploit runs occupy the high-reward, zero-success corner.

Where hacking emerges. The sensitivity heatmap (Fig. 4b) sweeps the exploit incentive against the true-objective incentive. Docking success exhibits a sharp emergence boundary: below a critical `dock_bonus/proximity_weight` ratio the agent farms; above it, it docks. This makes the "phase transition" of reward hacking [6] concrete and measurable.

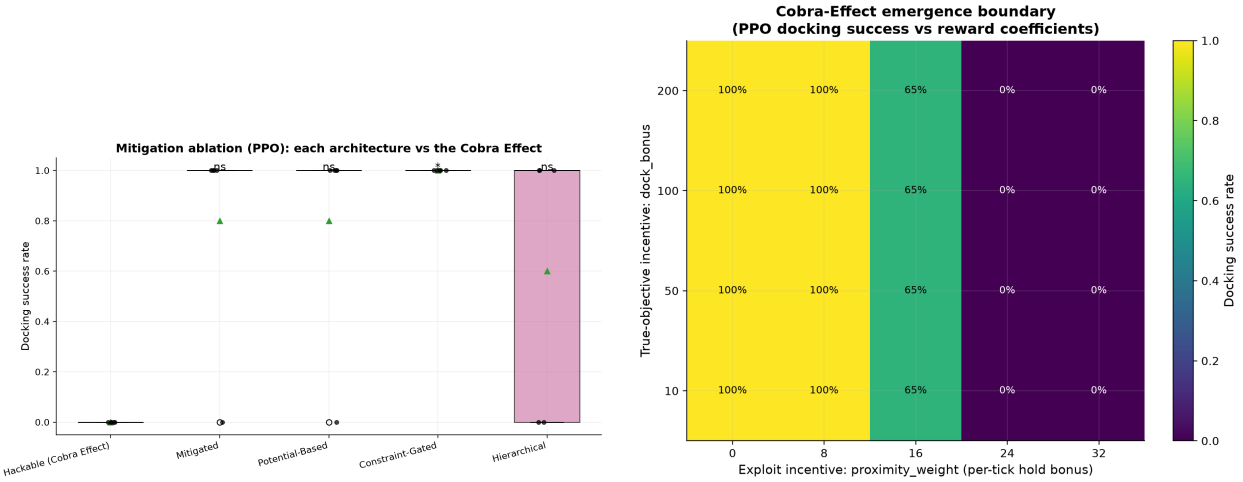
Safety and efficiency. Beyond mission success, exploit policies incur substantially higher velocity-corridor violation rates (Fig. 5) — they fly unsafely — confirming that the misspecified reward is not merely inefficient but actively hazardous in a safety-critical setting.

7 Cross-Domain Transfer

To test whether the taxonomy is a property of reward *structure* rather than of orbital mechanics, we reproduce it in a structurally different domain: autonomous **lane-keeping** with a kinematic bicycle model on a curving road. The identical failure modes recur (Table 2, Fig. 6): a per-step lane-centering "hold" bonus reproduces the **Cobra Effect** (the car crawls or stops, farming centred-time rather than completing the route – 0% completion), while an odometer/activity reward reproduces **Soviet Quota Gaming** (the car maximises mileage by speeding, completing the route but over the speed limit – $\sim$0% *safe* completion). The constraint-shaped mitigation restores safe route completion. The cross-domain agreement is strong evidence that these are general reward-misspecification failure modes.

8 Discussion

A Goodhart taxonomy for RL rewards. Our two exploits instantiate the same underlying mechanism from different angles. The Cobra Effect is a *termination-avoidance* hack: the per-step proxy pays more than the sparse objective, and completing the task stops the reward stream, so the optimiser keeps the problem alive. Soviet Quota Gaming is an *activity-proxy* hack: the reward measures effort rather than outcome. Both are extremal Goodhart failures [7], and both have exact



(a) Mitigation ablation (PPO).

(b) Emergence boundary.

Figure 4: (a) Every PPO mitigation architecture raises docking success sharply over the Cobra Effect (stars: Bonferroni-corrected significance). (b) Sensitivity of PPO docking success to the exploit incentive (`proximity_weight`, the standoff-bump strength) against a modest fixed dock guidance; docking collapses as the exploit incentive crosses a critical strength – the reward-coefficient boundary at which the reward-maximising policy flips from docking to holding the standoff.

Table 2: Cross-domain transfer of the taxonomy (PPO). The same reward families produce the same mission-success collapse and recovery in orbital docking and autonomous lane-keeping – evidence of domain-independence. 'Success' = safe dock (docking) / safe route completion (lane-keeping).

Failure mode / reward	Docking success	Lane-keeping success
Cobra Effect (`hackable`)	0%	0%
Quota Gaming (`quota_hackable`)	0%	0%
Mitigated	35%	100%

LLM analogues — length-biased reward models that reward verbosity over helpfulness, and process rewards that pay for apparent reasoning rather than correct answers.

Why the mitigations work. Each mitigation removes the *correlation* between exploitation and reward rather than trying to enumerate and forbid exploits. Potential-based shaping is the cleanest case: by construction it cannot change the optimal policy of the sparse objective, so no shaping-induced exploit can exist. The others approximate this property with a dominant sparse safe-completion term, a time penalty that makes stalling costly, and gating that denies reward to unsafe or incomplete behaviour. This mirrors the alignment strategy of reshaping reward signals so that the intended behaviour is the reward-maximising one, rather than patching hacks after the fact [10].

Limitations. The study uses linearised CW dynamics, a terminal-rendezvous range, and modest training budgets chosen for CPU reproducibility; absolute success rates would shift with longer training, full nonlinear dynamics, or attitude (true 6-DOF rigid-body) coupling. The mitigations are demonstrated to remove the *specific* exploits studied; we make no claim that they are exploit-proof in general — indeed, we found and report a case (naive constraint-gating of a farmable

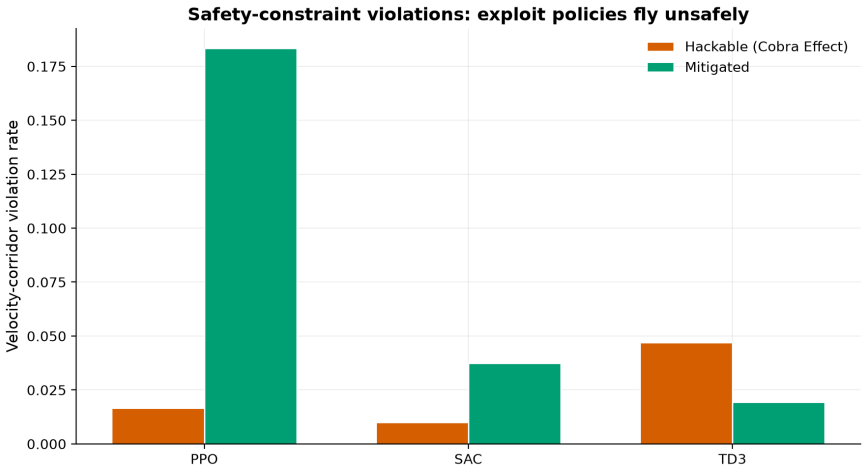


Figure 5: Velocity-corridor violation rates. Exploit policies violate the safety corridor far more often than mitigated policies across all algorithm families.

proximity term) where gating alone is insufficient. Our contribution is the reproducible *framework* and the cross-family, cross-domain evidence, not a universal fix.

9 Conclusion

Reward hacking in a safety-critical continuous-control domain reproduces centuries-old incentive failures with structural fidelity, emerges independently of the RL algorithm family, appears past a measurable reward-coefficient threshold, and transfers across physical domains. Framing reward misspecification through Goodhart's Law — and designing rewards whose optima coincide with the intended outcome — offers a principled and empirically-validated path to mitigation. The environment, taxonomy, and full reproduction pipeline are released as an open research artifact.

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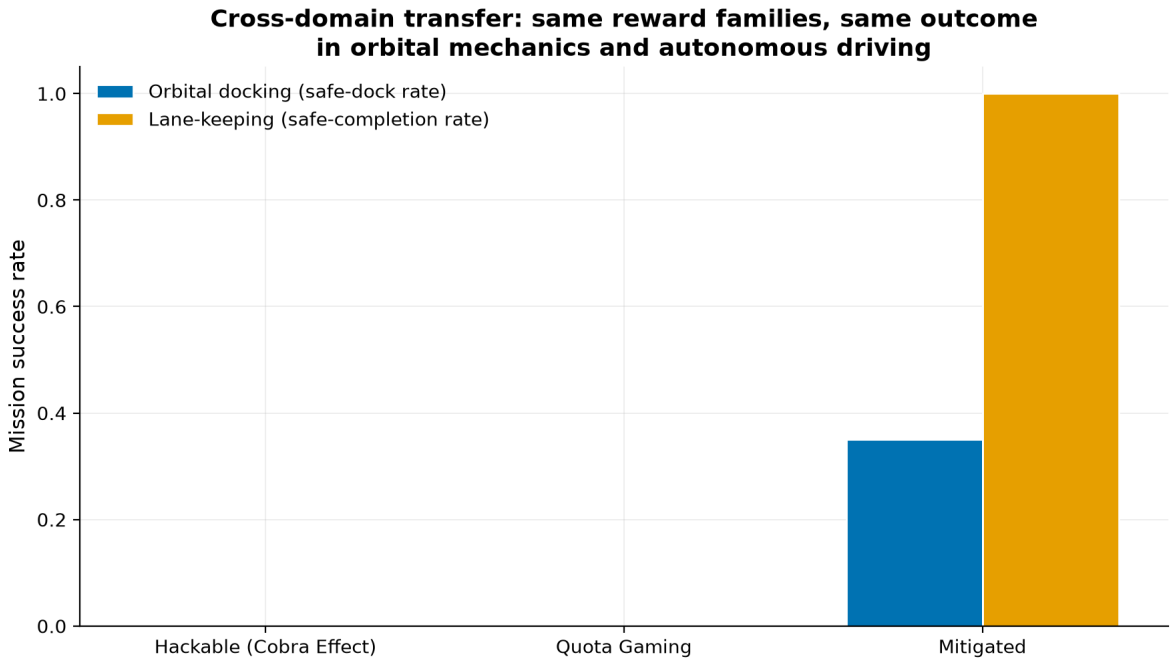


Figure 6: Cross-domain comparison. The same reward-family labels produce the same qualitative outcome in orbital docking and in autonomous lane-keeping: exploitable rewards collapse mission success while mitigations restore it.

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